

AVINASH AMUDALA

Software Engineer III

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SUMMARY

Software Engineer building production AI-agent platforms and distributed systems at Nokia. Currently leading an agentic QA framework — 57 skills, 9 orchestrators, 5 sub-agents, 25 tools — that delivered ~140× speed-up on test plan generation across 8 teams. Shipped a production RAG system over 800K+ documents and an open-source MCP server (60+ AI-callable tools). Published in IEEE and arXiv. Strong in Python, C++, Java, Large Language Models (LLMs), and cloud-native architectures (AWS, Kubernetes, Docker).

EXPERIENCE

Software Engineer

Aug 2024 - Present

Nokia Sunnyvale, CA

- Architected agentic QA framework (CMG Mobile Gateway): Designed and shipped an end-to-end agentic test automation system with 57 skills, 9 orchestrators, 5 sub-agents, and 25 tools that turns a single prompt ("complete testing for MG-XXXX") into per-team test plans, generated test scripts, regression runs, failure triage, auto-fix loops, and cross-repo dev handoff. Delivered ~140× speed-up on testplan generation (10 plans across 8 teams in ~1.7h vs ~30 engineer-days) over a 29K-node CLI tree, JIRA PRDs, and regression history.
- Built production RAG platform: Designed and deployed a retrieval-augmented generation system using FastAPI, Llama 3, FAISS, and BM25 to autogenerate test plans from 800K+ documents. Implemented hybrid vector + keyword search with re-ranking, achieving 4× faster retrieval and reducing manual test creation time by 60% across 3 engineering teams.
- Created open-source AI agent framework: Built MCP-Telecom, a Model Context Protocol server exposing 60+ AI-callable tools via SSH, NETCONF, SNMP, and gNMI. Published on PyPI. Features: safety layer (20+ blocked command patterns), config compliance engine, connection pooling, parallel execution, web dashboard, and Prometheus metrics.
- Engineered protocol simulation system: Built a packet-level replay engine using Scapy and Python async for automated testing of 5G core session flows. Designed distributed systems pipelines for packet capture and analytics processing 100K+ sessions.
- Shipped microservices and dashboards: Developed FastAPI + React + PostgreSQL services on Kubernetes with GitLab CI/CD. Built GraphQL and REST APIs for real-time monitoring, improving orchestration throughput by 40%.

Software Engineer

Jan 2023 - Aug 2024

Rochester Institute of Technology Rochester, NY

- Developed search platform using Solr + Express.js + GraphQL, improving query accuracy by 30% and response latency by 50% for 20K+ users.
- Built full-stack applications (Angular, Java, PHP) with AWS Lambda, S3, and Cognito for secure cloud-native access control. Optimized API caching, reducing p95 latency by 35%.
- Teaching Assistant for Intro to Software Engineering — mentored 60+ students on design patterns, testing strategies, and agile methodology.

PROJECTS & OPEN SOURCE

MCP-Telecom [Link](#)

Python, FastMCP, Netmiko, ncclient, pysnmp, gRPC, FastAPI, Docker

- First open-source MCP server for network infrastructure. AI agents (Claude, GPT) call 60+ tools to query devices, run compliance checks, discover topology, and execute parallel operations across 7 vendor platforms. Published on PyPI with full CI/CD, Docker, and Containerlab integration.

PROXIMA [Link](#)

Python, NumPy, SciPy, Pandas, Matplotlib, Bootstrap Statistics

- Research framework for validating proxy metrics in A/B testing. Composite reliability score combining effect correlation, directional accuracy, and segment-level fragility rate. Evaluated on 14M+ observations across 80 simulated experiments, 98.4% oracle agreement. Published on arXiv (2604.14352).

EasyApply AI 2024

Python, FastAPI, GPT-4, AWS Lambda, React

- AI-powered hiring platform using GPT-4 for resume parsing and candidate-job matching. Multi-modal extraction from PDFs and images. REST API on AWS Lambda. Reduced manual screening workload by 60%

PUBLICATIONS

IEEE: A. Amudala, E. Prabhu, "An IoT-Model for Monitoring Irrigated Crops" <https://ieeexplore.ieee.org/document/9885455>

IEEE ICESC 2022, pp. 1645–1651

arXiv: A. Amudala, "PROXIMA: A Reliability Scoring Framework for Proxy Metrics in Online Controlled Experiments" <https://arxiv.org/abs/2604.14352>

arXiv:2604.14352 [stat.ME], Apr 2026

EDUCATION

Rochester Institute of Technology

2022 - 2024

MS, Software Engineering — Coursework: Machine Learning, Cloud Systems (AWS), Data Structures & Algorithms, Database Design

Amrita Vishwa Vidyapeetham

2018 - 2022

BE, Electronics and Communication Engineering

TECHNICAL SKILLS

Languages Python, C++, Java, TypeScript, SQL, Bash, PHP, TCL

AI / ML Agentic frameworks (MCP, multi-agent orchestration, sub-agents, skill-based routing), RAG (FAISS, BM25, LangChain), LLMs (GPT-4, Claude, Llama 3), PyTorch, TensorFlow, Hugging Face, NLP (SpaCy, Transformers), A/B Testing, Agentic frameworks (MCP, multi-agent orchestration, sub-agents, skill-based routing), RAG (FAISS, BM25, LangChain), LLMs (GPT-4, Claude, Llama 3), Large Language Models (LLMs), ML infrastructure, NLP (SpaCy, Transformers)

Cloud & Infra AWS (EC2, S3, Lambda, SageMaker, RDS, Cognito), Docker, Kubernetes, Terraform, CI/CD (GitLab, GitHub Actions, Jenkins), Azure, AWS (EC2, S3, Lambda, SageMaker, RDS, Cognito), CI/CD (GitLab, GitHub Actions, Jenkins), distributed systems

Backend FastAPI, Node.js, GraphQL, REST APIs, PostgreSQL, Redis, Kafka, Solr, gRPC, Protocol Buffers, high-availability

Frontend React, Angular, TypeScript, HTML/CSS

CERTIFICATIONS & ACTIVITIES

Certifications: AWS Solutions Architect Associate • AWS Cloud Practitioner • Azure Fundamentals (AZ-900) • Nokia NRS I

Activities: Guest Lecturer — SJSU (AI, Robotics, 5G/Cloud), Santa Clara (Responsible AI). Peer Reviewer — AIAI 2026 (IFIP), ACM, DMLR, Elsevier, Springer Nature, MDPI